**10** **(a)** You are given that $(x^2+y^2)^3 = x^6+3x^4y^2+3x^2y^4+y^6$.

Hence, or otherwise, prove that $\sin^6\theta+\cos^6\theta = 1-\frac{3}{4}\sin^2 2\theta$ for all values of $\theta$. **[4]**

**(b)** Use the result from part **(a)** to determine the minimum value of $\sin^6\theta+\cos^6\theta$. **[2]**

**10(a)**

**10(b)**